

Boyi Yuan

Department of Economics
University of Calgary
437 Social Sciences Building
2500 University Dr NW
Calgary, AB, Canada T2N 1N4

Email: boyi.yuan@ucalgary.ca
Phone: +1 (778) 320-9160
Website: boyiyuanbob.github.io

Education

University of Calgary , Ph.D. Economics	2023- (expected 2029)
Committee: Erica Myers (Chair), Sarah Johnston, Benjamin Crost	
University of Sydney , M.A. Economics	2021–2023
Simon Fraser University , B.A. Economics	2016–2018
Fraser International College	2013–2015

Research Interests

Primary Fields: Energy and Environmental Economics, Applied Econometrics, Transportation
Secondary Fields: Behavioral Economics, Industrial Organization

Working Papers

Do Gasoline Price Changes Shift Mobility Choices? Evidence from German Walking Behavior

[Draft] Revise and Resubmit, *Journal of Environmental Economics and Management*

Abstract: Gasoline price policies are central to transportation decarbonization, yet little is known about whether higher fuel costs lead people to walk more. Using regional daily step data from consumer wearables in Germany between 2020 and 2022, I estimate how gasoline price changes affect walking, exploiting a shift-share instrument and event studies around two large 2022 price shocks. I find causal evidence that higher gasoline prices increase walking, plausibly reflecting substitution away from driving. The response is modest for routine price variation but substantially larger for policy-driven shocks, consistent with a role for price salience. Linking the walking response to the WHO HEAT model, I find that the health benefits of a policy-driven price increase are large relative to its direct emissions savings. The results suggest that fuel-price policies can raise walking along an overlooked margin and generate meaningful health co-benefits.

Selected Works in Progress

Do Pricing Algorithms Change Tax Pass-Through? Station-Level Evidence from Germany's Fuel-Tax Cut (with Ajornie Taylor)

Presented at the 60th CEA Annual Conference

Abstract: Do pricing algorithms affect the transmission of tax changes to consumers? We study this question using Germany's 2022 fuel-tax reduction (Tankrabatt) and subsequent reinstatement, which generated two large and opposing cost shocks. Combining minute-level gasoline prices for approximately 15,000 stations with a measure of algorithmic pricing adoption, we compare pricing responses between adopters and non-adopters. We find that algorithmic pricing does not affect the overall level of fuel-tax pass-through. However, adopters raise prices significantly faster following the tax reinstatement and reduce prices more slowly following the tax cut, amplifying the "rockets and feathers" pattern of asymmetric price adjustment. These results suggest that algorithmic pricing affects the timing of pass-through rather than its ultimate incidence, highlighting a previously unexplored channel through which artificial intelligence can influence consumer prices and policy transmission.

Strategic Response to Environmental Shocks: The Political Economy of Bank Branch Allocation in China (with Kangyu Qiu and Shukang Xiao)

Abstract: Environmental regulations can impose substantial adjustment costs on local industries. We study whether the state responds strategically to these costs through the spatial allocation of financial infrastructure. Using a comprehensive county-level panel of Chinese cities and a difference-in-differences framework, we examine how environmental regulation affects the placement of new bank branches. We find that regulation is associated with a significant increase in bank branches in affected areas, consistent with a targeted “defensive deployment” rather than purely market-driven expansion. However, greater physical access to banking does not translate into higher aggregate lending; instead, loans per capita decline. The results suggest that the state uses banking networks as a financial stabilizer, expanding access to finance to facilitate local adjustment to environmental regulation without necessarily increasing overall credit supply.

Riding for Free, Driving Less

Research Experience

University of Calgary

Research Assistant for Prof. Erica Myers

Jan 2025–present

Research Assistant for Prof. Alexander Whalley

Jul 2024–Dec 2024

Teaching Experience

University of Calgary

Teaching Assistant for ECON677: Environmental Economics (PhD/MA)

Winter & Fall 2026

Teaching Assistant for ECON707: Advanced Microeconomics Theory I (PhD)

Fall 2025

Guest Lecturer for PHIL345: Issues in Environmental Ethics

Summer 2025

Teaching Assistant for ECON395: Introduction to Econometrics

Summer 2025

Teaching Assistant for ECON715: Advanced Econometrics II (PhD)

Winter 2025

Teaching Assistant for ECON203: Principles of Macroeconomics

Winter 2024

Teaching Assistant for ECON303: Intermediate Economic Theory: Macro

Fall 2023

Grants, Fellowships & Awards

Doctoral Fellowship, *University of Calgary*

2023–present

Best Paper and Presentation Prize, *CREEA Early Career Workshop*

2026

Alberta Graduate Excellence Scholarship (×4), *Government of Alberta*

2023–2026

John S. Poyen Scholarship, *University of Calgary*

2026

Jim Gaisford Second Year Paper Prize, *University of Calgary*

2026

Honourable Mention, Best Student Paper Prize, *CREEA Annual Meeting*

2025

Graduate Faculty Council Scholarship – Doctoral, *University of Calgary*

2025

Microeconomics Prize, *highest score on Microeconomics Core Exam, University of Calgary*

2024

Econometrics Prize, *highest score on Econometrics Core Exam, University of Calgary*

2024

John S. Poyen Scholarship, *University of Calgary*

2024

Economics Alumni Graduate Scholarship, *University of Calgary*

2023

Conference & Seminar Presentations

2026 Electrify Society Summit (Montréal); Electricity Policy Bootcamp + Electricity Camp (Banff); 60th CEA Annual Conference (Vancouver); 12th CREEA Early Career Workshop (Vancouver); Berkeley/Sloan Summer School (Berkeley); 19th IAEE European Conference (Munich)*; AERE–SEA Annual Conference (Houston)*

2025 59th CEA Annual Conference (Montréal); 35th CREEA Annual Conference (Winnipeg); Electricity Lunch Seminar (UCalgary); Transportation Seminar (UCalgary)

2024 Environmental Economics Workshop (London, Canada)

* Scheduled presentation.

Professional Service

Journal Referee

Journal of Environmental Economics and Management

Conference Reviewer

AERE–SEA Annual Conference, 2026

Professional Memberships

Canadian Economics Association (CEA); Association of Environmental and Resource Economists (AERE); American Economic Association (AEA); Canadian Resource and Environmental Economics Association (CREEA)

Department Service

Economics Graduate Association (EGA) 2025–present
Vice President, Academic

EGA Reading Group Series, University of Calgary 2025–present
Founder/Organizer

Positions and Professional Experience

HD Education, Senior Tutor 2022–2024
Courses: ECON6002: Macroeconomic Analysis 1 (MA); ECMT6007: Analysis of Panel Data (MA), *University of Sydney*

YouPlus Education, Tutor 2017–2021
Courses: ECON260: Environmental Economics; ECON103: Principles of Microeconomics; ECON105: Principles of Macroeconomics, *Simon Fraser University*

Skills & Personal

Programming & Software Skills

ArcGIS; $\text{\LaTeX}$; Matlab; Python; R; Stata

Languages

Mandarin Chinese (native); English (fluent); Cantonese (intermediate); German (basic)

Citizenship

Canada

Last updated: September 2026